MECH 5200 – SEMESTER PROJECT DESCRIPTION: 2026 SPRING

VALENTIN BOUTROUCHE
(ADAPTED FROM THE SPRING 2025 PROJECT DESCRIPTION BY DAVID WILLIS)

1. Summary

You may work individually or in a group. Up to four (4) students form a group.

Each group/individual will perform a project that involves solving a PDE using one or more of the methods: FDM, FVM or FEM.

An analytical solution must be presented/derived if available. If not available, you should consider using commercial software (COMSOL, Ansys, or other) or a second PDE solution method to provide a benchmark solution.

Error vs. refinement should be presented for the L_∞ or L_2 norm as appropriate.

2. Suggested Project Components

The projects should aim to demonstrate a slightly higher level of complexity than the homework assignments in the class. Some of the ways to add complexity to your project could include less than three of the following components:

- **Increasing the geometric complexity of the problem being solved:** In this case you may wish to solve the PDE on a complex domain such as in cylindrical or spherical coordinates as opposed to Cartesian coordinates.
- **Examining systems of PDE equations:** Many physical systems are not simply one PDE, but rather they involve the solution of two (or more) PDEs simultaneously. This is the case in fluid mechanics with the conservation of mass and the conservation of momentum. Be careful however, that this complexity can quickly get out of control.
- **Solving non-linear problems:** Solving a non-linear PDE or system of PDEs is more complex and can be attempted by those comfortable with numerical methods looking for a challenge.
- **PDE based mesh generation:** There are several techniques that use PDEs to generate the meshes that are used later in the solution process. These meshes are usually of high quality and may be useful to somebody else's solution methodology.
- **Increasing the problem dimension:** For some of you, you may wish to try to solve the problems in class in a higher dimension than done in class. Eg. 1-D hyperbolic equations vs. 2D hyperbolic equations. You might even consider attempting to move to 3-Dimensions (FEM/FMD).
- **Implicit vs. explicit solutions in time:** You could explore the differences between explicit and implicit time marching.

- **Coupling PDEs with ODEs or algebraic constraints:** Some physical systems involve PDEs coupled with ordinary differential equations or algebraic relations, such as reaction kinetics coupled with diffusion or lumped-parameter models interacting with spatially distributed fields. This coupling introduces additional modeling and numerical complexity.
- **Advanced boundary and interface conditions:** Instead of simple Dirichlet or Neumann boundary conditions, projects may incorporate mixed (Robin) boundary conditions, time-dependent boundaries, or interface conditions between multiple materials with discontinuous properties.
- **Computational efficiency considerations:** Projects may investigate solver efficiency, including sparse matrix storage, iterative versus direct solvers, preconditioning strategies, or computational cost as a function of mesh resolution and time step size.

These are a few ideas on how to increase the complexity of your project problems. Feel free to either define your own project problem or consult with Dr. Boutrouche to define/pick a project.

3. Grading scale for complexity

The general grading scale for complexity is based on the following factors: whether the topic has been covered in the homework or lecture; when the topic was introduced in the semester; derivation or programming efforts needed to implement the method, linear vs. non-linear, dimension of the problem, regular/irregular domain, time-stepping method (as appropriate), etc.

Groups will be expected to solve problems of greater complexity than individual students.

4. Deliverables

(1) **Project proposal (10%)**: You should plan to submit your project proposal on Gradescope by Friday Feb. 13th. For group projects, a single submission linked to all members as well as a signed group contract.

This proposal (2 pages MAXIMUM) should be typed (single spaced) using a computer and will include – for some extra challenge, you may consider typing the project using LaTeX:

(a) **Summary:** A summary of the problem you wish to solve along with the PDE solution methods you will use.
(b) **Physical Problem:** A paragraph description including governing equations, boundary conditions and computational domains. One illustration is needed to specify the computational domain (Geometry) and boundary conditions.
(c) **Numerical Method:** You should write a paragraph summary of the numerical approach you will take to solve the problem. This should include any challenges you expect to encounter.
(d) **Validation**: A short summary of how you will validate your numerical solution to show you are getting the correct answer.

(e) **Approximate timeline:** For complicated problems, it is more realistic to start off with a simple version and add complexity gradually as you make initial progress. You should include a brief timeline of the development and testing schedule for your computer code.

(f) **References:** You should provide a list of references used in the development of the project proposal. This would include any use AI-LLM.

Depending on the quality and subject of the proposal, you may be asked to submit a revised proposal.

(2) **Computer code (20%)**: You must submit a working (MATLAB or similar) computer code that addresses the proposed problem. This is due at the end of the semester.

(3) **Demo Day (10%)**: You will attend demo day and showcase your project to the class in an end of the semester expo (last day of class – likely short class oral presentations).

(4) **Final technical report (60%)**: You must submit a typed technical report including the technical explanation of: (1) The physics of your problem, (2) The mathematics/numerical methods used , (3) The software/code designed and (4) validation of the results. A rubric will be provided.

The format will be similar to a journal article, and for the typical project should be approximately 6 pages (absolute max 8 pages) including results.

Example: https://iopscience.iop.org/article/10.1149/2.1011606jes/meta